

ROSHNI

The streetlight repair queue, reordered by who is actually walking.

THE CITY ALREADY KNOWS WHERE LIGHTS ARE REPORTED BROKEN

39,847

streetlight complaints in five and a half months of 2025 — **31.4% of every civic complaint Bengaluru filed**, and the largest single category in the city. Garbage is second, at 16,137.

There is no shortage of demand signal.



The repair system works. The order is wrong.

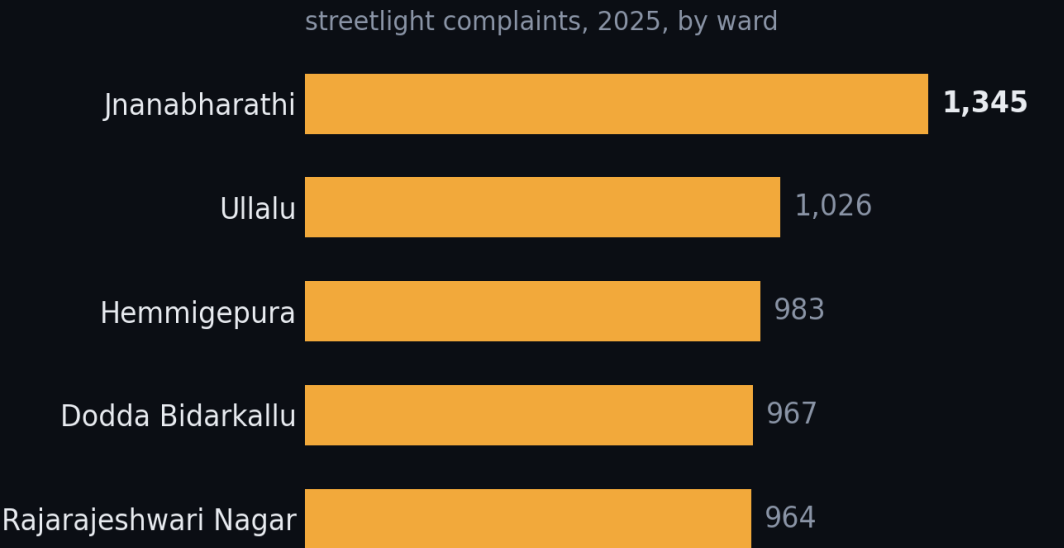
96.4% of electrical complaints closed



40,632 of 42,138. Road maintenance closes 58%. They are not slow and not under-resourced.

A complaint tells you someone noticed. It does not tell you how many people are exposed.

Complaining tracks who complains



Newer, peripheral, more car-owning wards file most. Older, denser, more-walked wards file fewer — not because their lights work.

Same register. Left: Grievance Status field. Right: Sub Category = "Street Light Not Working", grouped by Ward Name, 198 wards.

SO WE CHANGED THE UNIT OF MEASUREMENT

A lamp tells you what failed.

A road segment tells you who is exposed.

For every 200-metre stretch of road, Roshni estimates **darkness × pedestrian exposure × confidence**.

Darkness per segment is a quantity you can **estimate**. Lamp inventory is an enumeration you cannot **obtain** — and BBMP does not have one either. Their complaint records carry no coordinates and no pole IDs, so on day zero we are at parity with the municipality and ahead of it the moment a report carries a GPS fix.

11,029 segments · 1,180 km · 33 wards, across the locked ORR corridor.

One estimate. Three decisions.

THIS WEEK

What should the city repair?

A ranked worklist under a repair budget, scored by who is exposed rather than who complained.

Municipal queue · THE HERO

TONIGHT

Where should I walk?

Safer and shortest route side by side, always both. It refuses to path through ground we have no data for.

Citizen route

NEXT

Where is a report worth most?

The streets where a single answer would change a repair decision — high exposure, low confidence.

Citizen report

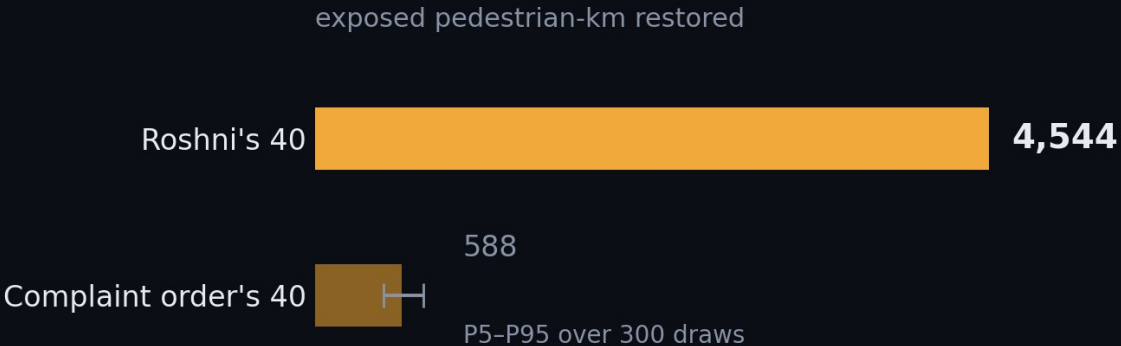
One variable, opposite signs. Footfall is exposure in the repair queue — more people at risk on a dark street, so fix it first. It is protection in the router — eyes on the street, so prefer it. Use one sign in both and your router calls an empty, brightly lit industrial road safe at one in the morning.

SAME CREW. SAME BUDGET. DIFFERENT ORDER.

7.7×

Under a 40-repair budget, Roshni's forty restore **4,544 exposed pedestrian-kilometres** against 588 for the complaint queue's first forty.

**The gain is not from being cleverer inside a ward.
It is from visiting the right wards.**



And the floor, volunteered **1.9×** against a steelman BBMP that somehow picks the single worst segment in every ward it visits — which a coordinate-free complaint log cannot do.

Three further stress tests — 4.9× against the luckiest of 300 draws, 6.4× restricted to wards where we hold lamp labels, 5.4× with every lamp label deleted — are in Appendix A.

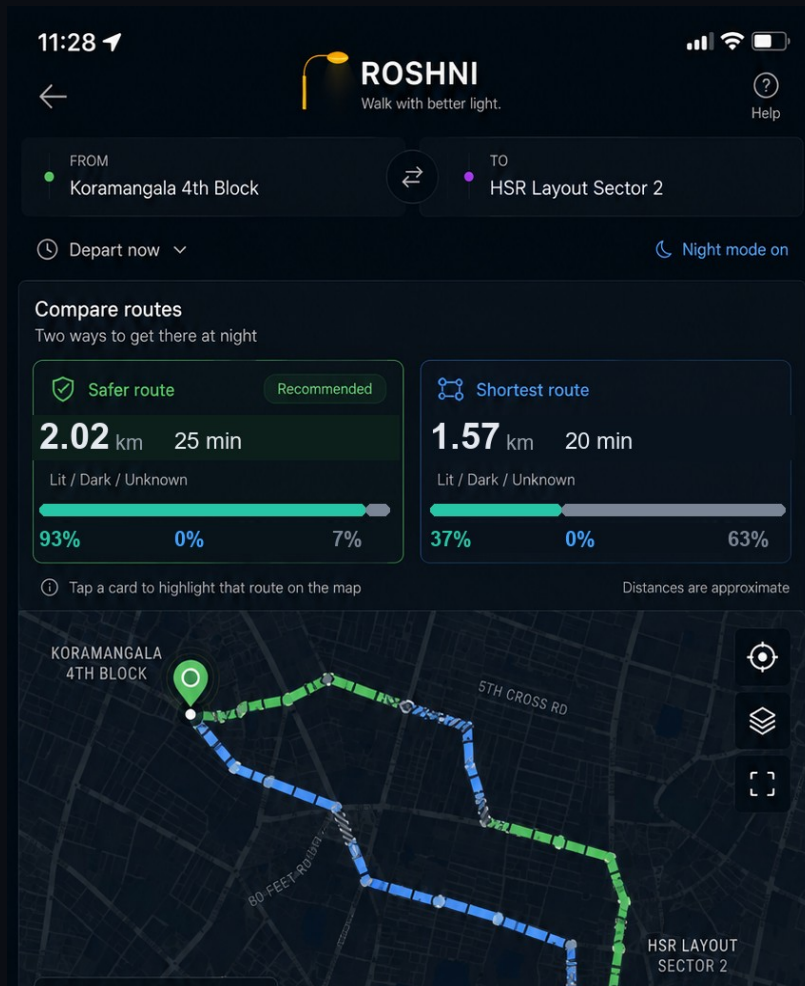
What the city sees



The ratio, live — the budget slider recomputes in the browser. **Municipal priorities** — the municipality's to set, not ours. **Confidence is marked** — low-confidence rows are hatched, not hidden.

Prototype, running client-side on the fixture export. No network call on the demo path.

What a walker sees



Shortest: 1,575 m, and 63% of it crosses ground we are not confident about.
Safer: 2,021 m, 7% unknown — it routes around 18 low-confidence segments.

Six minutes to walk on ground we actually have data for.

The shortest route is always shown beside the safer one. We do not auto-select and we do not hide it — the choice is the walker's, and we say plainly which stretches we have no data for.

But routing cannot fix the worst streets

Delete each of the 25 worst segments and look for another way between its endpoints:

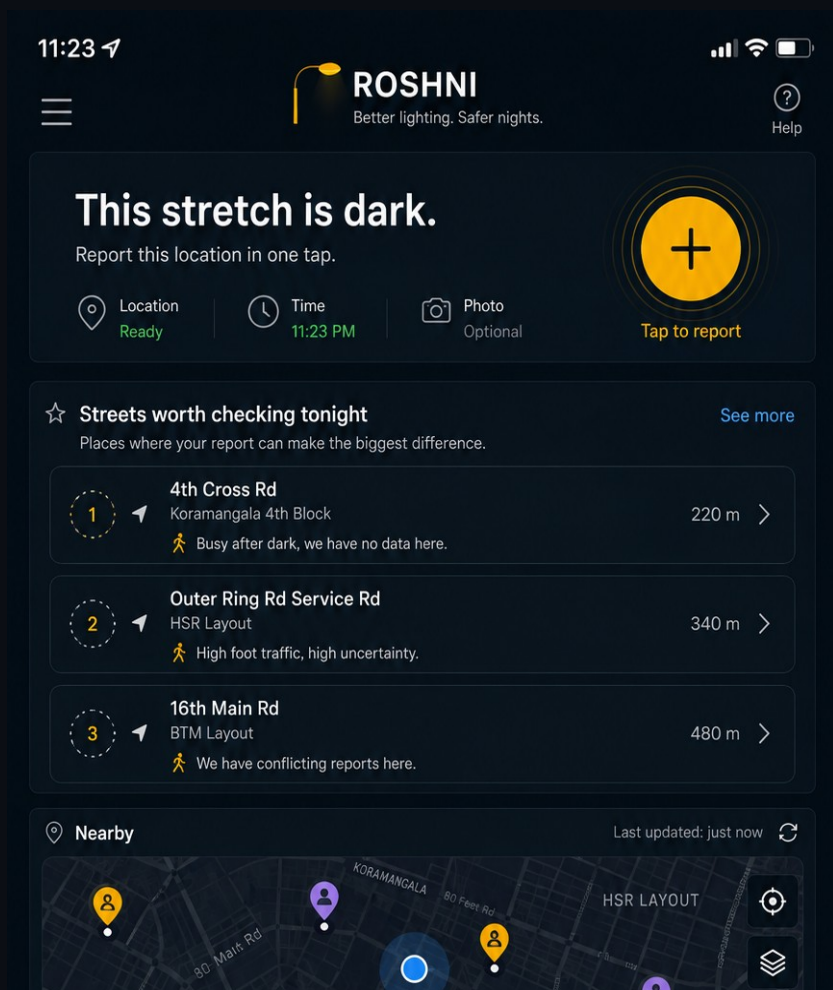
6 have no walkable alternative at all

19 have a bypass over 3× the segment's length

17.9× median bypass · worst 206.9×

On the streets that most need fixing there is nowhere else to walk. Only repair solves them.

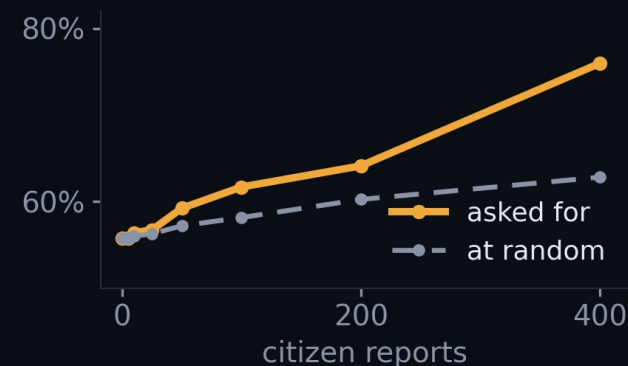
What a reporter sees



The empty state is not “no reports yet.” It is **“streets worth checking tonight”** — so the app directs reporting instead of waiting for it.

93.7% of our 11,029 segments have no lamp evidence at all and run on the ward prior alone. That is not a flaw we are hiding — it is why every segment carries a variance, and it is what this screen is for.

A report is worth most exactly where we are least sure and most people walk.

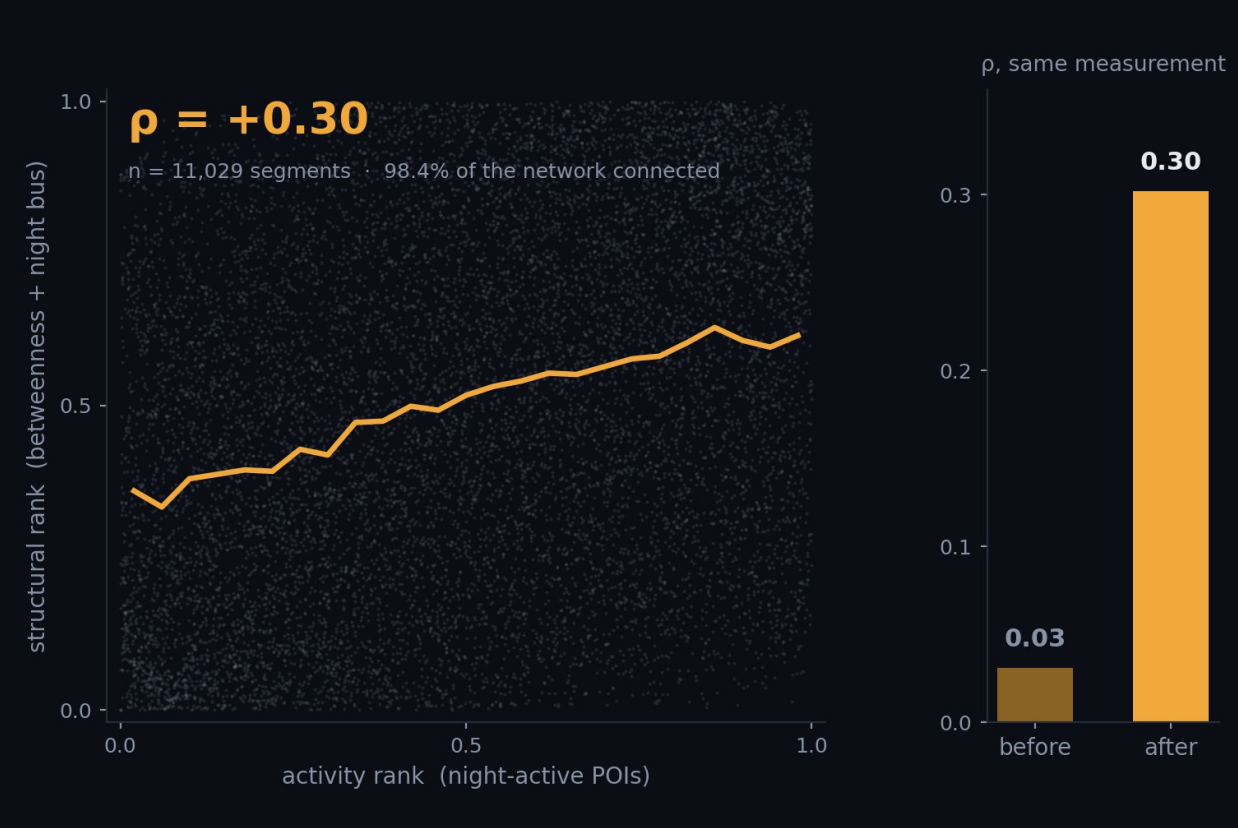


With zero reports the prior-only ranking already captures **56%** of what a perfectly-informed ranking would restore. 400 targeted reports takes it to **76%**; 400 random reports reach only 63%.

Simulation, not a field trial — labelled as one in out/curve.json.

We got a null, distrusted it, and found our own bug

Validation needs a signal you did not use to build the estimate. And you are validating an ordering, not a magnitude — “412 pedestrians an hour” is unfalsifiable; “these streets rank above those” is checkable.



The first answer was $\rho = 0.03$. Two proxies that should have agreed didn't. So we went looking, and found we were keying the walk graph on **rounded coordinates** — junctions where two roads genuinely meet weren't merging. The network had shattered, and the betweenness we were feeding the model was computed over 55% of the city.

Keyed it on OSM node ids and re-ran over eight times the area. 98.4% now connects, and two proxies built from completely unrelated inputs — night commerce, and pure network topology plus bus service — converge at $\rho = 0.30$.

Convergent	ρ +0.30, +0.44 vs night bus	
Criterion — night pedestrian crashes		not yet measured
Ground truth — stratified field counts		not yet measured

Measured, not chosen — and honest about the rest

WHAT WE MEASURED

The prior is calibrated, not picked

A citizen audit coded 469 Vasanthanagar lamps on two dates 15 days apart: 45.9% of dead lamps were repaired, 4.0% of working lamps died. That is a measured equilibrium of 8.0% dead — so the prior comes from a failure rate we can point at.

Two disjoint proxies agree

$\rho = +0.30$ between night commerce and pure network topology, and $+0.44$ against night bus departures. Unrelated proxies do not agree at 0.30 by accident.

The win survives deleting our own best data

Restricted to the wards where we hold lamp labels: 6.4×. With every lamp label deleted and the ranking built on ward prior and exposure alone: 5.4×.

WHAT WE GET WRONG

93.7% of segments run on a ward prior alone

690 of 11,029 carry any lamp evidence. Every other one carries a confidence band that says so — the router avoids the least-certain fifth, the report screen targets it. But most of the map is an estimate.

Our labelled lamps are geographically concentrated

OSM's working tags come from citizen surveys, and those surveys walked some wards and not others. 657 broken to 471 working is a property of who went mapping, not Bengaluru's outage rate.

Two inputs we simply do not have

OSM records opening_hours for 6.7% of POIs here, so we weight night activity by POI class instead. And tree canopy is a large effect on how lit a Bengaluru street feels, with no data at all.

The data exists. It's one approval away.

93 Indian cities publish streetlight counts through MoHUA's Smart Cities portal. Zero publish coordinates — the D24 template is a KPI return, not an asset register.

The pole-level geometry is on IUDX, for our city, published by the Electronics City Industrial Township Authority, **marked Private** — behind a Request Resource button. The sample record is public:

```
{"deviceId": "L0302",  
  "location": {"type": "Point", "coordinates": [77.666046, 12.841758]}}
```

Roshni's lamp record is that record, copied verbatim. **“We plug into the municipal feed the day access is granted” is a schema fact, not a promise.** When it opens, the prior is replaced by measurement and the bands collapse — and the queue, the exposure model and all three screens are unchanged.

THE ASK

Approve the IUDX request.

We are not asking anyone to survey anything new. The register already exists.

7.7×

Same crew. Same budget. Different order.

Appendix A — every number, and where it came from

126,974	civic complaints, 1 Jan – 19 Jun 2025	1,508 / 1,128	lamps in snapshot / labelled working=yes no
39,847 · 31.4%	of them “Street Light Not Working”	690 · 6.3%	segments with any lamp evidence
96.4%	electrical closure rate (40,632 / 42,138)	27.6%	mean P(dark) across all segments
58.0%	road-maintenance closure rate	ρ +0.30 / +0.44	activity vs structural / vs night bus
421,113	streetlights across 198 BBMP wards	405 · 6,187	BMTC stops / departures after 21:00
11,029	segments scored · 1,180 km · 33 wards	3,417 / 9,199	night-active POIs of all POIs
95,557	walk-graph nodes · 98.4% one component	17.9×	median bypass, 25 worst segments

Counterfactual at a 40-repair budget: 4,544 vs 588 exposed pedestrian-km = 7.7×. Luckiest of 300 draws 4.9×. Steelman 1.9×. Surveyed wards only 6.4×. All lamp labels deleted 5.4×.

All figures out/stats.json, out/curve.json, out/routes.json — pipeline run 22 Aug 2026.

Appendix B — the machinery

THE DARKNESS MODEL

```
P(lit)    = coverage × 0.920
           coverage = ward lamps/km ÷ 33.3
           33.3 = 30 m standard pole spacing
           0.920 = measured working rate
Beta(α₀, β₀) = (K·P(lit), K·(1-P(lit)))    K = 2
```

Evidence updates α and β , each decayed on a 30-day half-life so a confirmed repair genuinely clears a segment:

IUDX zero current after dusk	2.0 → β
repair confirmed by a reporter	1.5 → α
OSM working = no / = yes	1.0 → β / α
way lit = no	0.8 → β
citizen report, corroborated	1.0 → β
OSM lamp, no working tag	0.4 → α

Every element exists in data/pipeline.py. No ML model, no cloud layer, no auth service. VIIRS nightlights are 500 m per pixel and are not used to judge a street.

THE PIPELINE

Inputs

OSM walk graph (95,557 nodes, 98.4% connected) · BMTC GTFS (6,187 night departures) · night-active POIs (3,417 of 9,199) · ward lamp density (421,113 lamps, 198 wards) · OSM working tags (1,128 labelled) · citizen reports · IUDX telemetry, gated

Two models

EXPOSURE $E(\text{segment})$ = activity + structural, kept separately so convergent validation is possible. DARKNESS $\text{Beta}(\alpha, \beta)$, mean = $P(\text{dark})$, var = confidence.

Four outputs

repair_priority = $E \cdot P(\text{dark}) \cdot \text{confidence}$, solved as a knapsack under budget. route_penalty = $L \cdot (1 + \lambda d \cdot P(\text{dark}) + \lambda e/E + \lambda u \cdot \text{var})$ — note E divides. Confidence band drives the routing refusal. Report targeting on an upper confidence bound.

Vasanthanagar transition matrix

469 lamps, 17 Jun → 2 Jul 2019. DEAD → DEAD 33, DEAD → OK 28, OK → DEAD 15, OK → OK 360. 45.9% repaired, 4.0% failed, equilibrium dead share 8.0%.